

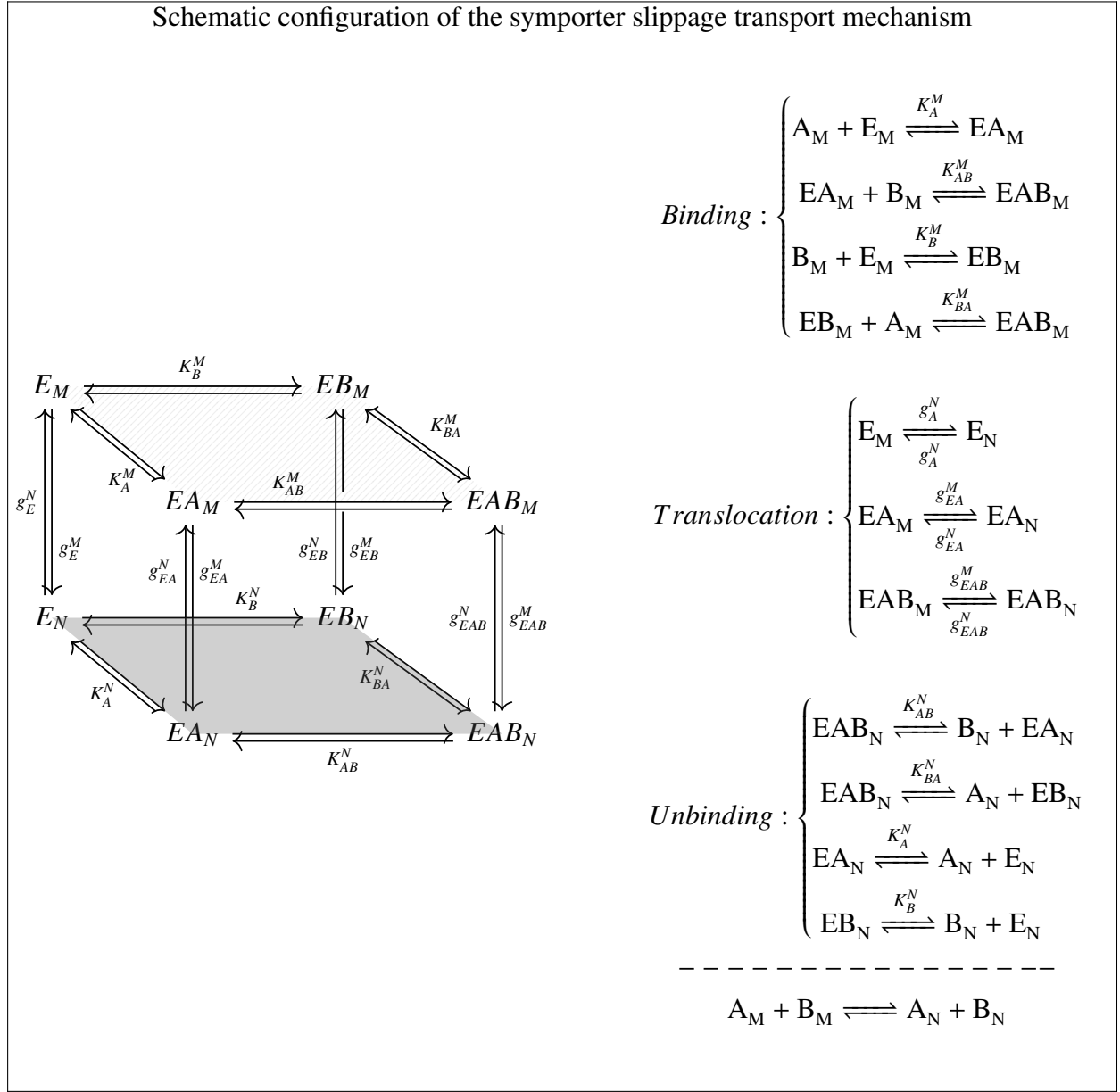


Figure 11: A schematic configuration of the symport slippage transport mechanism and the regarding net transport flux. Dissociation constants (K_A^M , K_{BA}^M , K_B^M) and the translocation rate constants (g_E^M , g_{EA}^M , g_{EB}^M , g_{EAB}^M) are depicted in the reaction cycle in the left panel. $[E]_t$ is the total number of transporter per unit area of the membrane. In this model empty carrier and the partially loaded carrier can slipp across the membrane. The flux equation is transformed to the general form of the resistance from Stein model [131] (part 1/2 continued on the next page).

Net Transport Flux from M to N

$$J_{A, \text{symporter}}^{M, N(\text{net})} = [E]_t \left(\frac{(g_{EA}^M \alpha^M + g_{EAB}^M \alpha'^M \beta^M)(g_E^N + g_{EB}^N \beta^N) - (g_{EA}^N \alpha^N + g_{EAB}^N \alpha'^N \beta^N)(g_E^M + g_{EB}^M \beta^M)}{R_M R_{NN} + R_N R_{MM}} \right) \quad (55a)$$

$$J_{B, \text{symporter}}^{M, N(\text{net})} = [E]_t \left(\frac{(g_{EB}^M \beta^M + g_{EAB}^M \alpha'^M \beta^M)(g_E^N + g_{EA}^N \alpha^N) - (g_{EB}^N \beta^N + g_{EAB}^N \alpha'^N \beta^N)(g_E^M + g_{EA}^M \alpha^M)}{R_M R_{NN} + R_N R_{MM}} \right) \quad (55b)$$

Where

$$[E]_t = [E]_M + [EB]_M + [EA]_M + [EAB]_M + [EAB]_N + [EA]_N + [EB]_N + [E]_N$$

$$\begin{aligned} \alpha^M &= \frac{[A]_M}{K_A^M}, \alpha'^M = \frac{[A]_M}{K_{BA}^M}, \beta^M = \frac{[B]_M}{K_B^M} & \alpha^N &= \frac{[A]_N}{K_A^N}, \alpha'^N = \frac{[A]_N}{K_{BA}^N}, \beta^N = \frac{[B]_N}{K_B^N} \\ R_M &= (1 + \alpha^M + \beta^M + \alpha'^M \beta^M) & R_N &= (1 + \alpha^N + \beta^N + \alpha'^N \beta^N) \\ R_{MM} &= g_E^M + g_{EA}^M \alpha^M + g_{EB}^M \beta^M + g_{EAB}^M \alpha'^M \beta^M & R_{NN} &= g_E^N + g_{EA}^N \alpha^N + g_{EB}^N \beta^N + g_{EAB}^N \alpha'^N \beta^N \end{aligned}$$

Figure 11: A schematic configuration of the symport slippage transport mechanism and the regarding net flux transport equation. Dissociation constants (K_A^M, K_{BA}^M, K_B^M) and the translocation rate constants ($g_E^M, g_{EA}^M, g_{EB}^M, g_{EAB}^M$) are depicted in the reaction cycle in the left panel. $[E]_t$ is the total number of transporter per unit area of the membrane. In this model empty carrier and the partially loaded carrier can slipp across the membrane. The flux equation is transformed to the general form of the resistance from Stein model [131] (part 2/2 continued from previous page).

Two-substrate symporter ordered binding models:

Kinetic schemes for sequential binding (ordered binding) cotransport mechanisms involving two solutes are depicted in Figures 12 and 13. When A binds first, the symport transport mechanism shown in Figure 12 applies, and the corresponding net flux of A and B through the symporter can be expressed using equations 56b and 56c, respectively. Similarly, the mechanism for the case in which B binds first is illustrated in Figure 13, and the corresponding net outward flux of A and B can be obtained using equations 57b and 57c, respectively. In these equations, $J_{\text{symporter}}^{M, N(\text{net})}$ is the steady-state turnover rate of the symporter (expressed in equations 56a and 57a, accordingly), and the parameters α, β , and R are related to the carrier translocation rate constants and the dissociation constants, which are expressed in terms of experimentally measurable parameters at the bottom of the Figures 12 and 13, accordingly. These models were developed and parameterized by Stein (2012) [131]. In these models, it is assumed that the partially loaded carriers (EA and EB) cannot